

FeedbackNow FlexBox v1.2 FRD

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0.1 1. Overview

FlexBox v1.1 is a battery-operated LoRaWAN feedback device designed for indoor deployment (e.g., bathrooms, retail spaces, airports). The device captures user sentiment via 6 physical buttons and supports employee check-in/check-out workflows via NFC. An optional e-paper display shows timestamps and status. All data is uplinked (device → network) to a LoRaWAN network for backend processing; downlinks (network → device) deliver commands and configuration.

System Concept:

The device serves two user types: (1) **Public users** (customers, visitors) who press a single button to submit feedback, and (2) **Staff** (custodians, managers) who use NFC cards to check-in, check-out, or log issues. Each interaction generates an uplink message containing a timestamp, event type, and relevant data (button ID, counter, or User ID). The backend receives this raw data and applies business logic (session tracking, analytics, alerts). The device itself is stateless, it captures and transmits data without interpreting meaning.

Hardware Components:

Component	Description
MCU	NRF52840
Buttons	6 physical buttons (0-5)
NFC	PN5180 module, ISO15693 protocol
LoRa	SX1262 module
Display	E-Paper / EPD (optional, Variant A only)
Timekeeping	External RTC (Real-Time Clock, maintains time when powered off)
Storage	EEPROM (non-volatile memory); SDHC card slot (out of scope for v1.2; reserved for future)
Feedback	1 LED
Power	Replaceable battery (9600mAh), ADC monitoring
Unused	Reed switch (reserved for future)

Note: Values marked with an asterisk (*) throughout this document are configurable and subject to customer sign-off.

0.2 2. User Roles & Interactions

0.2.1 2.1 User Types

User Type	Description	Authentication
Public	General foot traffic (customers, visitors)	None
Staff	Custodians, managers, employees	NFC card (ISO15693, 4-byte User ID)

0.2.2 2.2 Button Layout

Button	Color	Public Mode	Staff Mode
0	Green	Positive feedback (Happy)	Check-in
1	Yellow	Neutral feedback (Satisfied)	Check-out
2	Red	Issue (per decal)	Registered Vote
3	Red	Issue (per decal)	Registered Vote
4	Red	Issue (per decal)	Registered Vote
5	Red	Issue (per decal)	Registered Vote

0.3 3. System States & Transitions

0.3.1 3.1 Button Combinations

Combo	Hold Duration	Action
Single button (0-5)	Tap	Public feedback vote
Button 0+1	2 seconds*	Enter Staff Mode
Button 0+1+2 (in Staff Mode)	3 seconds*	Trigger LoRa join (deliberate / first-time)
Button 0+1+2+3 (in Staff Mode)	10 seconds*	Reboot device
Button 0+1+5	3 seconds*	Device Info display

0.3.2 3.2 State Machine

0.3.3 3.3 System Modes & States

State	Entry Trigger	Timeout	On Timeout	LED
Normal	Boot complete / other states exit	None	-	Off
Staff Mode	Button 0+1 hold 2s*	10s*	Return to Normal	Solid on
NFC Scan	Button press in Staff Mode	5s*	Return to Normal	Blink 1Hz
Device Info	Button 0+1+5 hold 3s*	30s*	Return to Normal	-
Reboot	All 6 buttons hold 10s*	-	-	Solid 3s

0.3.4 3.4 Timeout Summary

Context	Timeout	Behavior
Staff Mode waiting	10 seconds*	Return to Normal (no uplink)
NFC Scan waiting	5 seconds*	Return to Normal (no uplink)
Device Info display	30 seconds*	Return to Normal

Context	Timeout	Behavior
Public button lockout	5 seconds*	Accept next press
EPD “Thanks for your Feedback!”	5 seconds*	Return to “Last Cleaned”

0.3.5 3.5 Interaction Summary

Action	User	Trigger	Result
Feedback Vote	Public	Single button press	LED confirmation, LoRa uplink
Check-in	Staff	Staff Mode → Button 0 → NFC tap	EPD: “Cleaning in progress”, NFC uplink per schema
Check-out	Staff	Staff Mode → Button 1 → NFC tap	EPD: “Last Cleaned” + timestamp, NFC uplink per schema
Registered Vote	Staff	Staff Mode → Button 2-5 → NFC tap	Uplink with button ID + NFC uplink per schema

0.4 4. Functional Requirements

0.4.1 4.1 Button Behavior

Public Mode (Normal):

- Single button tap triggers feedback vote
- LED solid for 1 second* confirms acceptance
- 5-second* lockout prevents spam (silent rejection, no LED)
- Uplink Event 0x00 with button ID and running counter

Staff Mode:

- Button press in Staff Mode selects action type and starts NFC scan
- No lockout between actions (staff may need multiple votes)

Button Counters:

- Each button maintains independent running counter (24-bit, max 16,777,216)
- Counters persist across power cycles (stored in EEPROM)
- Counters persist across battery replacement

Disconnected Operation:

- Button presses accepted regardless of network connection state
- When disconnected: Only button counters persist in EEPROM (no full event queue)
- LED and EPD feedback operate normally (user sees no difference)
- On rejoin, latest button counts sent via Event 0x07 (Counter Sync) so backend can reconcile

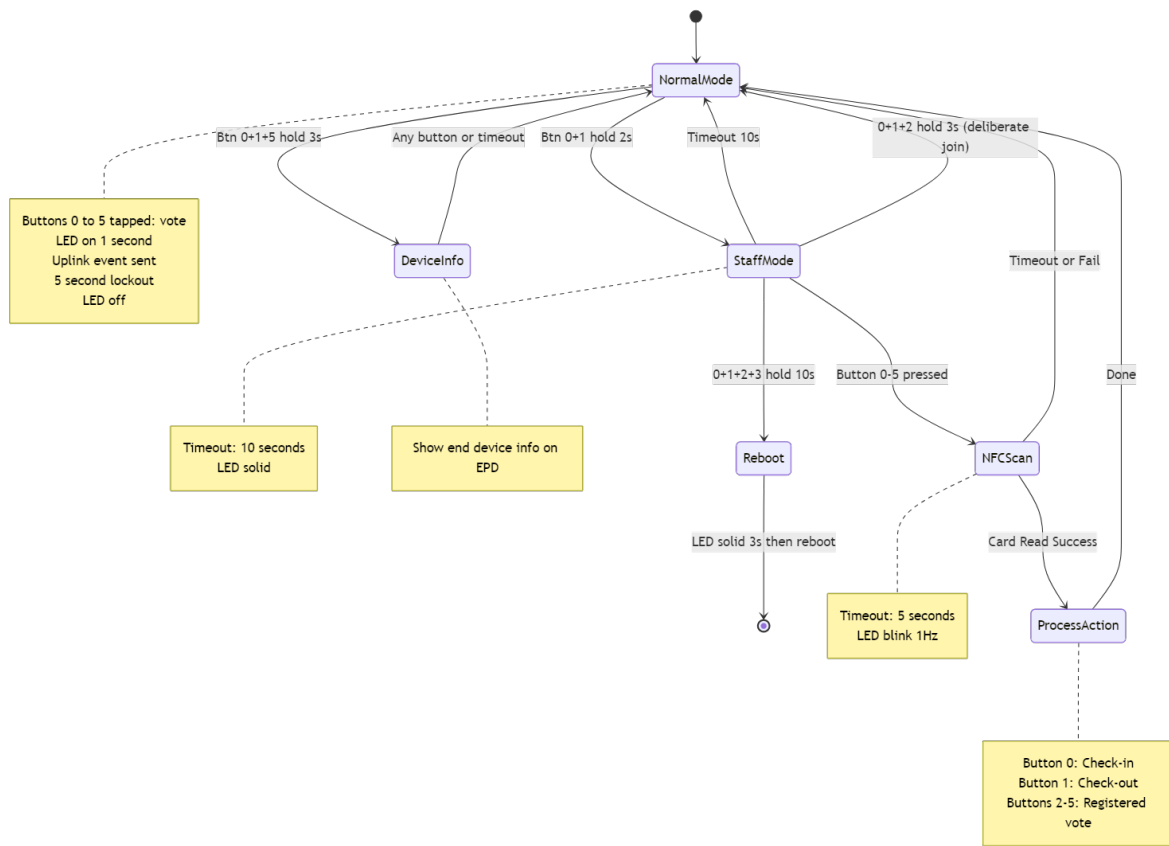


Figure 1: State Machine

0.4.2 4.2 NFC / Staff Mode

Staff Mode Entry:

- Button 0+1 held for 2 seconds*
- LED turns solid to indicate active state
- 10-second* timeout to select action

NFC Scan Flow:

1. Staff presses button (0-5) in Staff Mode
2. LED blinks 1Hz (waiting for card)
3. Staff taps ISO15693 card within 5 seconds*
4. Device reads 4-byte User ID from designated memory block
5. Action executes based on button pressed

Actions:

Button	Action	Event Type	EPD Update
0	Check-in	0x01	“Cleaning in progress”
1	Check-out	0x02	“Last Cleaned” + timestamp
2-5	Registered Vote	0x03	No change

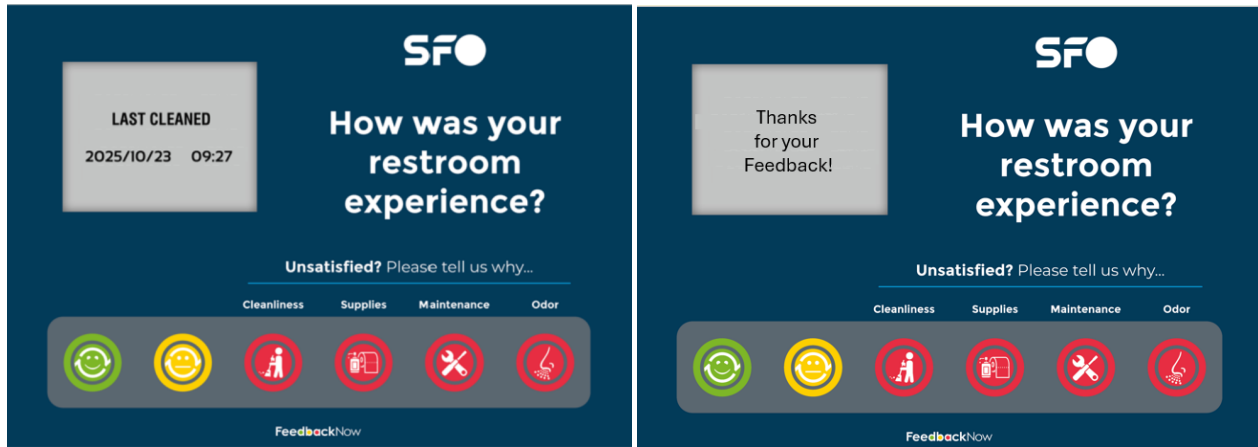
NFC Failure:

- Read timeout or error: LED 3 fast blinks, return to Normal

0.4.3 4.3 E-Paper Display (Variant A Only)

Display States:

State	Content	Trigger
Default	“Last Cleaned” (heading) + “yyyy/mm/dd hh:mm” (timestamp)	Boot, Check-out
Feedback Ack	“Thanks for your Feedback!”	Public button press
Cleaning	“Cleaning in progress”	Check-in
Device Info	DevEUI, FW version, battery %, counters	Button 0+1+5 hold
First boot (awaiting join)	“Ready” or similar	Before first deliberate join
Boot / Join	“Connecting...”	During LoRa join attempt



Timing:

- “Thanks for your Feedback!” displays for 5 seconds*, then returns to default
- “Cleaning in progress” persists until check-out. Public buttons are not blocked during cleaning; votes are always allowed (C1).
- If custodian does not check out, EPD auto-reverts after 45 minutes* to “Last Cleaned” with current (RTC) timestamp (C2).

Downlink EPD Updates:

- Backend can push “Last Cleaned” timestamp via downlink
- EPD downlink updates are processed right after the user vote flow: after “Thanks for your Feedback!” a 5-second* gap is applied, then any pending EPD downlink (e.g. “Last Cleaned”) is applied. This batches downlink with the user uplink and avoids overwriting the thank-you message or delaying updates until the next heartbeat.

0.4.4 4.4 LED Feedback

State	LED Pattern	Duration
Power On	2 quick flashes	~200ms
Join Success	3 quick flashes	~300ms
Normal / Idle	Off	-
Button Accepted (public)	Solid on	1 second
Button Rejected (lockout)	Off	-
Staff Mode Active	Solid on	Until exit
NFC Waiting	Blink 1Hz	Until scan/timeout
NFC Read Fail	3 fast blinks	~600ms
Check-in Confirmed	3 blinks	~2 seconds
Check-out Confirmed	3 blinks	~2 seconds
Registered Vote Confirmed	3 blinks	~2 seconds
Reboot Triggered	Solid on → off	3 seconds → reboot

0.4.5 4.5 LoRa Communication

Network Configuration:

- LoRaWAN Class A
- OTAA activation
- US915 frequency plan
- EU868 frequency plan
- Unconfirmed Uplinks

Uplink Event Types:

Event Type	Code	Trigger	Payload Contains
Button Press	0x00	Public vote	Timestamp, Button ID, Counter
Check-in	0x01	Staff NFC	Timestamp, User ID
Check-out	0x02	Staff NFC	Timestamp, User ID
Registered Vote	0x03	Staff NFC	Timestamp, User ID, Button ID
Heartbeat/Status	0x04	Daily timer	Timestamp, Battery %, Status
Low Battery	0x05	Threshold crossed	Timestamp, Battery %
Counter Sync (Rejoin)	0x07	On successful rejoin after disconnect	Button ID, Counter (per-button uplink; backend reconciles from last known count vs post-rejoin count)

Downlink Commands:

Command	Code	Description
Update EPD	0x01	Set “Last Cleaned” timestamp
EPD Refresh	0x02	Full e-paper refresh (clear ghosting)
Status Request	0x03	Trigger status uplink (Same as Heartbeat)
Reset Counters	0x04	Reset button counters to zero

Heartbeat:

- Frequency: Once daily*
- Timing: DevEUI-based jitter (DevEUI = unique 64-bit device identifier; jitter distributes load across 24 hours)
- Purpose: Battery status, RTC sync opportunity, downlink delivery window

Disconnected Operation (Counters Only):

- When network unavailable, only button counters are persisted in EEPROM (no full event payload queue)
- On rejoin: Firmware sends uplinks for each button with latest counter (Event 0x07 Counter Sync: Button ID + Counter)
- Backend reconciles from last known count vs post-rejoin count to infer votes that occurred during disconnect
- Counter-sync uplinks use rate limiting to avoid network congestion

Network Reconnection:

- Device attempts rejoin every 1 hour* when disconnected
- On successful join: Counter-sync routine runs (per-button Event 0x07 uplinks with latest counts)

0.4.6 4.6 Storage

EEPROM:

- Button counters (6×24 -bit)
- LoRa DevNonce (replay protection)
- Network state flag (connected/disconnected)
- First-join flag (“has__joined__once”; used for deliberate first-join procedure)
- CRC validation on read
- No full event log is stored; only counters and state are persisted

SDHC Card (out of scope v1.2):

- SDHC card slot is not used in this version. Planned for a future version: daily log files (button counts, events), counter backup, optional operation without card, Event 0x06 on low space. Retained here for version control.

Data Persistence:

- Counters written to EEPROM on each button press

0.4.7 4.7 Power Management

Battery:

- Replaceable, non-rechargeable
- Capacity: 9600mAh (1-2 cells)

Low Battery:

- ADC monitors battery voltage
- Below threshold: Uplink Event 0x05
- No local LED indication (rely on backend alerts)

Sleep Modes:

- Deep sleep between events (button GPIO + RTC alarm wake)
- Light sleep during NFC scan timeout

0.4.8 4.8 Boot Sequence

Note: Device enters Normal Mode regardless of network state. If disconnected, device operates normally and retries connection every 1 hour*.

0.4.9 4.9 Housekeeping

Housekeeping tasks run during daily heartbeat:

Task	Description
RTC Resync	DeviceTimeReq MAC command (correct drift) - if connected
Battery Sample	ADC reading for heartbeat payload
EEPROM → SDHC Flush	Persist counters to SD card
Counter Validation	CRC check, repair from backup if needed
SDHC Health	Check remaining space; if below threshold*, uplink Event 0x06

Task	Description
EPD Refresh	Full refresh every 7 days* (clear ghosting)

Periodic Rejoin (Separate from Housekeeping):

Task	Interval	Description
Network Rejoin	Every 1 hour* (when disconnected)	Attempt OTAA join if disconnected
Queue Flush	On reconnect	Transmit queued events after successful join

0.5 5. Firmware Variants

Two firmware variants are built from the same codebase with compile-time flags.

0.5.1 5.1 Variant Comparison

Feature	Variant A (with EPD)	Variant B (without EPD)
E-Paper Display	Yes	No
Button Feedback	LED + EPD text	LED only
“Last Cleaned” display	Shown on EPD	Not applicable
“Cleaning in progress”	LED + EPD text	LED pattern only
“Thank you” message	LED + EPD text	LED only
Device Info display	Shown on EPD	Status uplink triggered
NFC / Staff Mode	Full support	Full support
Check-in/out	Full support	Full support (no timestamp display)
Downlink EPD update	Supported	Command ignored
All other features	Identical	Identical

0.5.2 5.2 Build Configuration

Single codebase with `CONFIG_EPD_ENABLED` flag:

- `CONFIG_EPD_ENABLED=y` → Variant A
- `CONFIG_EPD_ENABLED=n` → Variant B

0.6 6. Error Handling

0.6.1 6.1 Error Handling

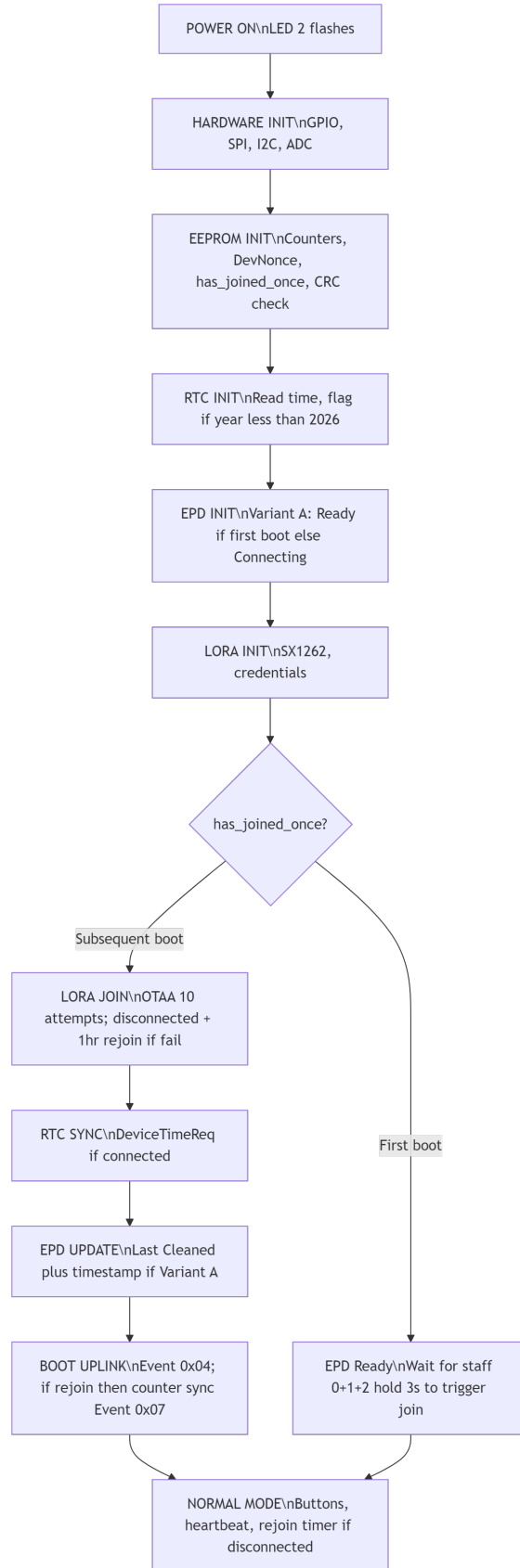


Figure 2: Boot Sequence

Error	Detection	Behavior	Recovery
LoRa Join Failure	No JoinAccept after attempt	Retry with exponential backoff	Enter disconnected mode, retry every 1 hour*
Uplink enqueue/transmission failure (local)	No local ACK	Queue event for retry	Retry on next uplink opportunity
NFC Read Failure	Timeout or read error	LED: 3 fast blinks	Return to Normal Mode
NFC Read Timeout	5s* with no card tap	LED off	Return to Normal Mode
EEPROM CRC Failure	CRC mismatch on read	Use defaults, set first-boot flag	No backup in v1.2 (SDHC out of scope)
RTC Invalid	Year < 2026	Flag for sync	Sync via DeviceTimeReq after join
Low Battery	ADC below threshold	Uplink Event 0x05	Continue until power loss

0.6.2 6.2 Network Recovery

Key Principle: Device NEVER permanently shuts down due to network failure. It remains fully operational in disconnected mode and continuously attempts to reconnect.

0.6.3 6.3 Network Resilience Summary

Design Philosophy: The device is designed for unattended deployment with no field technician access. It must gracefully handle network outages and automatically recover.

Disconnected Mode Capabilities:

Feature	Behavior When Disconnected
Button presses	Accepted normally, LED feedback works
EPD updates	Works normally (uses local RTC)
NFC check-in/out	Works normally; counters persist
Counters	Increment and persist in EEPROM
User experience	Identical to connected operation

Rejoin Counter Sync:

Parameter	Value
Trigger	Successful network rejoin
Mechanism	Per-button Event 0x07 uplinks with latest counter
Backend	Reconciles from last known count vs post-rejoin count
Rate	Rate-limited to avoid congestion

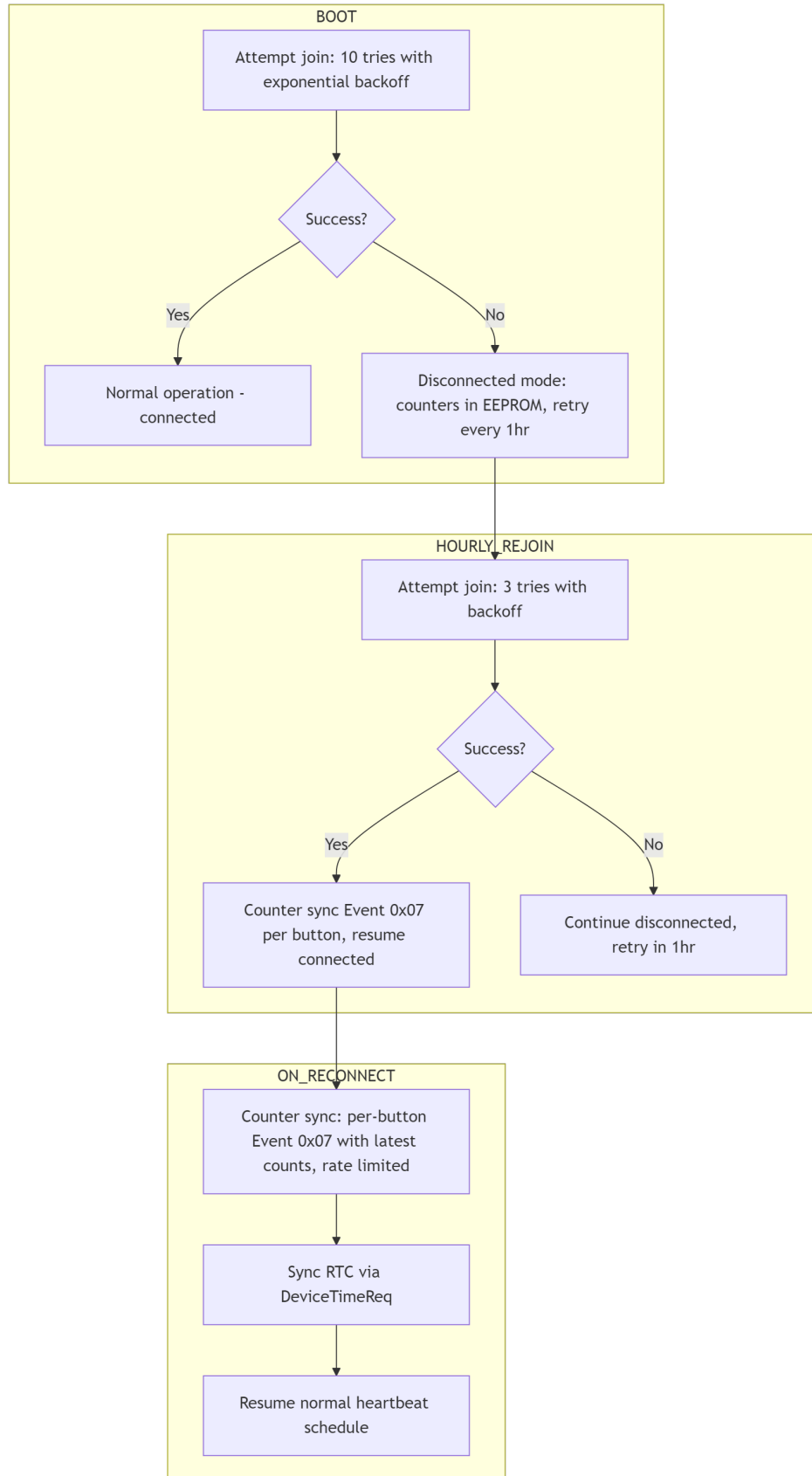


Figure 3: Graceful Network Recovery

Reconnection Strategy:

Trigger	Retry Behavior
First boot (no has__joined__once)	No join attempts; wait for staff-mode + 0+1+2 deliberate join combo
Subsequent boot (disconnected)	10 attempts* with exponential backoff, then disconnected mode
Hourly timer	3 attempts* with backoff
After reconnect	Immediate counter sync (Event 0x07 per button) + RTC sync

Network State Diagram:

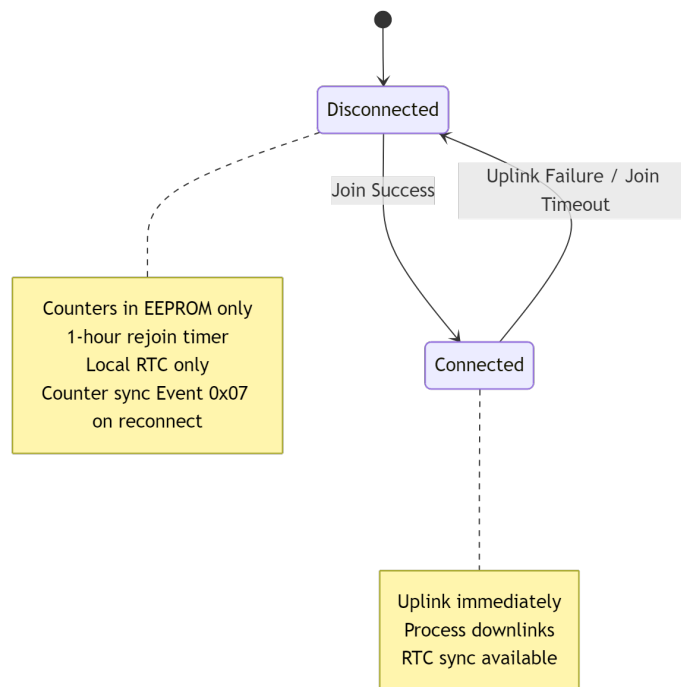


Figure 4: Network State Diagram

0.7 7. Clarifications Required

#	Item	Description	Options	Resolved
C1	“Cleaning in progress” button blocking	Behavior of public buttons while cleaning is in progress	- A) Block public buttons during cleaning?- B) Allow votes during cleaning?	B) Allow votes during cleaning – always allow.
C2	Custodian no-checkout timeout	How EPD behaves if custodian does not check out	- A) EPD stays “Cleaning in progress” indefinitely- B) Auto-revert after X hours- C) Auto-revert to previous timestamp- D) Auto-revert to current time	B) Auto-revert after 45 minutes*; D) Auto-revert to current time.
C3	Downlink EPD latency	Potential delays if EPD updates are tied to the daily heartbeat	- Updates may be delayed up to 24 hours.- Downlinked updates might not be truly “on demand.”	Resolved: EPD downlink applied after user vote (thank you + 5s* gap), batched with user uplink – no 24h delay.

0.8 8. Summary

0.8.1 8.1 Summary Of Design Decisions

#	Item	Decision
A1	Button IDs	0-5 (6 total)
A2	Staff Mode entry	Button 0+1 hold 2 seconds*
A3	Device Info entry	Button 0+1+5 hold 3 seconds*
A4	Reboot trigger	Enter Staff Mode, then 0+1+2+3 hold 10 seconds*
A5	NFC data format	4-byte User ID from designated memory block
A6	NFC card type	ISO15693
A7	Device is stateless	Backend handles session logic (check-in/out pairing)
A8	Public lockout	5 seconds* between presses
A9	Staff lockout	None (multiple actions allowed)
A10	Heartbeat frequency	Once daily*
A11	Heartbeat timing	DevEUI-based jitter (distributed across 24 hours)

#	Item	Decision
A12	RTC sync	DeviceTimeReq MAC command after join
A13	Initial “Last Cleaned”	RTC time at first boot
A14	Counter persistence	Survives power cycle and battery replacement
A15	Counter size	24-bit per button (max 16,777,216)
A16	Reed switch	Hardware present, unused in v1.2
A17	Single LED	All UI feedback via one LED
A18	LoRaWAN Class	Class A
A19	Activation	OTAA
A20	Join / first boot	First boot: no auto-join; deliberate join via Staff Mode + 0+1+2. After has_ <u>joined</u> _once: 10 attempts* on boot, then retry every 1 hour* (never shutdown)
A21	Disconnected operation	Device fully functional when disconnected; only counters persisted in EEPROM
A22	Rejoin sync	On rejoin, per-button Event 0x07 (Counter Sync) uplinks with latest counts; backend reconciles from counter deltas
A23	Network resilience	Device never permanently shuts down due to network failure
A24	SDHC	Out of scope v1.2; reserved for future (Event 0x06, logs, backup)

0.9 Document History

Document Version: 1.2

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Author: Jatan J. Pandya

Version	Date	Changes
1.0	January 2026	Initial draft

Version	Date	Changes
1.2	February 2026	Reboot: Staff Mode + 0+1+2+3 hold 10s; <i>deliberate join</i> : <i>Staff Mode + 0+1+2 hold 3s</i> . EPD downlink after thank you + 5s* gap. LED: check-in/check-out/registered vote 3 blinks. Disconnected model: counters only; Event 0x07 Counter Sync on rejoin. SDHC out of scope v1.2. First-boot no auto-join; has_joined_once flag. Renames: uplink failure (local), RTC < 2026, disconnected mode. Section 7 resolved (C1–C3).